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GNN-RL Agent-Assisted Tourism Planning for Electric Vehicle Travelers

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***Abstract*—The tourism industry has unleashed a keen interest in undergoing a major shift from conventional fossil fuels to renewable energy sources. However, electric vehicles (EVs) could not travel long distances due to depleted batteries and infrequent locations of renewable-specific electric charging stations. Additionally, EV-specific solutions are prone to considering several factors such as elevated roadways, traffic conditions, driver profiles, and so forth that impact travel decisions while identifying an optimal travel plan. This article proposes a novel approach of identifying an optimal travel plan by empowering Graph Neural Networks (GNNs) and Reinforcement Learning (RL) agents. Additionally, it includes a Particle Swarm Optimizer (PSO) to optimize the attained solutions. The proposed approach adapts to the dynamic travel conditions of EVs; it attempts to predict the energy consumption and time required to reach the destination in real-time. Experiments were carried out to identify travel paths and directions by feeding location maps. The experimental results highlighted that the proposed system was able to find an optimal path 79% of the time with a mean reward of 170. The proposed GNN-RL was integrated in a map so that travelers could utilize them at large.**

***Keywords*-Electric vehicles, Graph Neural Networks, Particle Swarm Optimizer, Reinforcement Learning.**

I. Introduction

Rising global warming remains a huge concern for humans and climate change-specific decision-makers due to increasing sea levels and temperature values. Reports show that fossil fuels contribute to carbon emissions, reaching over 78 percent [9].

Accordingly, tourism-specific transportation industries of various countries, including developing countries, have volunteered to utilize non-fossil fuel-based vehicles, aiming at preventing ecological damage to tourism locations – i.e., the sale of electric vehicles (EVs) increased [16].

Utilizing electric vehicles in the tourism sector, however, encounters challenges while identifying optimal travel paths. They are listed as follows:

1) EV-related Concerns – i.e., EVs could not travel long distances due to limited charging capabilities; they deplete electric charges proportionate with speed; and the batteries of EVs are expensive;
2) Non-Availability – the charging stations of EVs are not frequently found in ecological tourism locations. Also, the electric power sources are limited in several countries when compared to the required energy demands.
3) Non-real-time Solutions – Most available solutions are either inconsistent or static ones. In fact, the roads and travel routes are highly dynamic, including the battery conditions of EVs.

This article proposed a GNN-RL agent-assisted tourism planning mechanism for EV travelers. It utilized reinforcement learning agents to dynamically choose an optimal travel route considering various factors, including the battery conditions of vehicles.

Experiments were carried out at the IoT Cloud research laboratory to study the importance of applying GNN-RL agents. To get the EV-specific travel information in real time, sensor-specific data of EVs are necessary. However, to mimic the similar sensor-based data of EVs in real time, a Simulation of Urban Mobility (SUMO) tool [11] was utilized in experiments. SUMO was capable of loading custom maps from OpenStreetMap and creating random travel trips for EVs.

Experiments were carried out pointing to the tourism locations in Kochi, Kerala, India. Out of 100 different experiments, our proposed system was capable of finding more accurately the next steps in a travel path, considering battery conditions and travel conditions.

The remaining part of the article is organized as follows: Section II describes the existing methods and previous research works in the context of electric vehicle routing. Section III details the methodology used in the study; Section IV provides the experimental results; and, Section V briefly describes the conclusion and future work.

II. Related Works

EVs have increased in recent years, particularly in the tourism sector, due to evolving climate change conditions. Consequently, the research relating to EVs has significantly improved in various dimensions – i.e., researchers have attempted to perform a) battery optimization, b) placement of charging stations [15], c) routing EVs [18], and so forth.

Initially, researchers developed optimization strategies without the assistance of AI-specific algorithms to identify travel routes of EVs [4]. Notably, the authors of [21] proposed an

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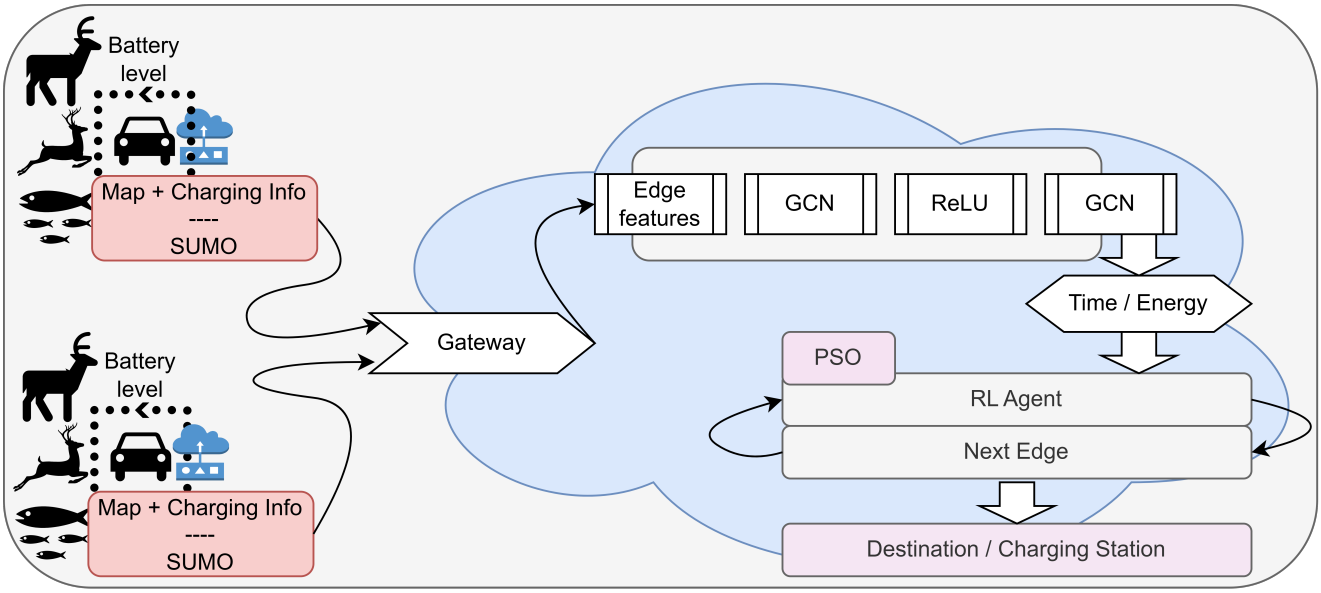


Fig. 1: Proposed GNN-RL-based Tourism Planning System

improved ant colony optimization and adaptive large neighborhood search-based bi-level algorithm for the resolvability of routing and charging feasibility of electric vehicle-based travels; the authors of [6] attempted to solve the EV routing problem using a Niched Pareto genetic-based heuristic algorithm.

In [8], the authors introduced time windows using ant colony optimization by penalizing the time taken in identifying the travel routes of EVs; the authors in [19] introduced a smart transportation model using a metaheuristic hybrid algorithm for the variable routing problem with time windows.

In the meantime, a few authors attempted to minimize the energy consumption of batteries of EVs using optimization algorithms. For instance, Joon Moon et al. [5] focused on integrating a recharge-specific scheduling algorithm into travel routes. The authors attempted to address challenges such as limited charging infrastructure, intermittent access, and temporal factors such as charger waiting times based on availability, charging durations, and fluctuating traffic conditions.

Qihenng Xu in [12] modeled the route planning problem as mixed integer programming and used a Pareto ranking-based genetic algorithm for placing charging stations, considering charging cost, time window, environment, and charging station construction cost. In [14], a suggestion for an energy-optimal prediction algorithm was studied while opting for a specific route of travel to air quality. However, this work did not include the integration of EV charging stations in real-time.

Several researchers have studied the importance of involving AI- and sensor-assisted technologies in the EV routing problem of the tourism sector. Notably, the researchers of [13] applied a fuzzy-based optimization algorithm to route buses in rural tourism; the authors of [17] examined the electric vehicle routing problem by explicitly modeling battery, electric vehicles' speed, and charging capabilities under stochastic traffic conditions; and Dariush Ebrahami et al. [3] integrated reinforcement learning to learn complicated scenarios, including battery consumption patterns.

The authors of [2] proposed a surrogate accelerated evolutionary algorithm. The main part of this algorithm had a transformer network-based surrogate model designed for predicting total driving distance for customer-exclusive routes.

Most of the AI-assisted EV-specific routing solutions have been studied in urban environments. Notably, Kai Wu, Fei Hao et al. proposed a federated graph convolution network in [7]. However, these authors did not involve a reinforcement learning algorithm or similar AI algorithms to consider a larger set of constraints.

In recent years, the application of reinforcement learning in EVs for solving various EV-related problems, including routing problems, has increased. Notably, the authors of [1] have utilized deep reinforcement learning agents to detect travel paths considering battery and charging constraints; the authors of [20] have compared the deep reinforcement learning algorithms in hybrid EVs. However, these works have not included the application of the routing problem in the tourism sector. Also, it could be observed that the agents had to learn a lot of information before suggesting travel routes of EVs.

Among the available research works, there are very limited works that utilize GNN to observe the travel location-based embeddings and RL agents to diligently plan the travel routes of EVs in real-time considering several complicated parameters, including battery lifetime, of the tourism sector. This article applied GNN- and PSO-assisted RL agents to identify travel paths in tourism locations.

III. GNN-RL-based Tourism Planning

Identifying travel routes and considering complicated decision parameters, such as energy consumption of batteries and

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distance of charging stations, are some of the important steps in EVs. This section details the proposed tourism planning methodology, involving GNN, PSO, and RL agents, to detect an optimal travel plan for tourism travelers. A pictorial representation of the proposed tourism planning system is shown in Figure 1.

The core components of the proposed GNN-RL-based tourism planning system include a) a GNN for modeling energy and travel time patterns, b) optimizers, and c) RL agents. The functional details of these components are described in the following subsections.

A. GNNs

The information of travel routes needs to be captured and represented as graph-structured data nodes in GNNs. Each travel node can be formulated as a feature vector of length, number of lanes, centroid_x, and centroid_y of tourism locations. The leading path is considered as edges in GNNs.

In the proposed work, a two-layer GraphSAGE Convolutional Network (GCN) is created to learn the embeddings of real-world locations – i.e., the energy consumption and travel time patterns of tourism-related roads. Algorithm 1 represents the working principle of the proposed GNN.

As noticed in algorithm 1, initially, the features of nodes and the possible edges are identified. Next, the energy consumption of batteries of the identified edges is predicted using GNNs. The predicted time and energy values are fed to RL agents for further identification of the next possible path (i.e., edge).

Algorithm 1 Edge Regression with Geometric GNN

Require: Node features x, edge index $edge_index$, edge features $edge_attr$, edge targets $targets$
Ensure: Predicted edge energies $\hat{y}$
Create Geometric Data Object:
Initialize `Data` object D with x, $edge_index$, $edge_attr$, $targets$
Define EdgeEnergyGNN model:
Define neural network layers:
- `conv1` : GCNConv($node_feat_dim$, 16)
- `conv2` : GCNConv(16, 16)
- `edge_mlp` : MLP($2\times16+edge_feat_dim \rightarrow 32 \rightarrow 1$)

Forward pass:
  $h \leftarrow \mathrm{ReLU}(\texttt{conv1}(x, edge_index))$
  $h \leftarrow \mathrm{ReLU}(\texttt{conv2}(h, edge_index))$
  For each edge (i, j):
    Concatenate h_i, h_j, $edge_attr_{ij}$ as input to `edge_mlp`
    Predict edge energy $\hat{y}_{ij}$ using `edge_mlp`
Training Loop:
Set device to GPU if available, else CPU
Initialize `EdgeEnergyGNN` and optimizer (Adam, $lr = 0.01$)
Move data and model to device
for $epoch = 1$ to N **do**
  Set model to training mode
  Zero optimizer gradients
  $\hat{y} \leftarrow \mathrm{model}(x, edge_index, edge_attr)$
  $loss \leftarrow \mathrm{MSE}(\hat{y}, targets)$
  Back propagate $loss$
  Update model parameters with optimizer
  if $epoch \bmod 20 = 0$ **or** $epoch = N$ **then**
    Print $epoch$ and $loss$
  end if
end for
Predict Edge Energies:
Set model to evaluation mode
$\hat{y} \leftarrow \mathrm{model}(x, edge_index, edge_attr)$ with no gradients
return Predicted edge energies $\hat{y}$

B. PSO Optimizer

The PSO optimizer is included in the proposed tourism planning system to apply particle swarm optimization principles similar to other optimization algorithms – i.e., the genetic algorithm, the Grey Wolf, and so forth.

In the PSO optimizer, the next optimal charging station is decided if the RL agent decides to identify the heading point. To do so, the optimizer includes several complicating parameters such as power, price, distance, lane, and travel queue to identify the costs of a particular traveling station.

The travel station cost is calculated using different weights set on parameters. For instance, the following weights are set up in the use case: "$distance$" : 0.30, "$lane$" : 0.15, "$queue$" : 0.25, "$price$" : 0.15, and "$power_inv$" : 0.15. The idea is to minimize distance, queues, and price; and to maximize power for faster recharge of EVs. With these objectives and weights, the PSO optimizer is set up to tune the algorithm while identifying the next charging station in a tourism plan. In PSO, each particle attempts to find the best charging station. Later, the global best is observed from the individual best particles.

C. RL Agents

To implement incremental optimal path finding under dynamic battery constraints, a reinforcement learning (RL) model was designed. In the proposed approach, the RL agents incorporated the output received from the GNN and particle swarm optimizer to generate an optimal path for an electric vehicle from source to destination, including charging batteries during the travel plan. The RL agent takes decisions dynamically to follow a path by continuously interacting with the environment.

The RL algorithm includes the following key terms: a) State: It is defined as the current node and the current battery level of the vehicle; b) Action space: K+1, where k represents the station and k+1 as the next scheduled charging station; c) Reward function: Reaching the destination is considered the best solution and is rewarded the most. But it is also penalized if the battery gets depleted.

The RL agent was designed in a way that it returns a set

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of values such as observation, reward, terminated, truncated, and information on different occasions to learn the environments using rewards – i.e., a) 150, if the vehicle reaches the destination; b) 50, if the vehicle reaches a charging station; c) -50, if the decision gets a timeout; and d) -10.0, if the vehicle reaches the dead end.

In short, the proposed work combined GNN, PSO, and RL agents to predict the optimal route for electric vehicles.

IV. Experimental Results

This section explains the findings of our proposed method in various dimensions as follows: a) Energy and travel time prediction, b) Success rate and RL rewards, c) Trip duration and impacts, d) Battery behavior analysis, and e) Map integration.

A. Experimental Setup

To input travel locations, an OpenStreetMap tool was utilized. It converted the travel spots and charging station locations to digraphs and represented the locations in an XML format.

Additionally, a tourism city map of Kochi, Kerala, was loaded into the SUMO environment. Upon loading the tourism city map, SUMO provided a few sets of output files as follows: a) battery-output.xml – it contains information about per vehicle energy consumption over time; b) fcd-output.xml – it involves the vehicle position and lane occupancy information; c) Kochi.net.xml – it includes the edges, nodes, lane definitions, lengths, and IDs; and d) edge-data.xml – it contains the aggregated traffic and performance metrics information.

TABLE I: Predicted Energy Consumption and Travel Time of Travel Locations (Edges)

Edge ID	Energy (kWh)	Travel Time (s)
-1000000000	60.245323	6.8603177
-10042159	13.202025	1.0236499
-100000000	735.2673	47.463852
-10052196	71.71851	5.284247

The output files, represented in XML format, were unified, and a dataset was created that quantifies the per-edge energy consumption – i.e., the energy consumption if a next travel trip were chosen. The dataset, therefore, is dynamic real-time information directly fed from the travel map of the city. It consists of several features such as edgeID, sampledSeconds, travelTime, waitingTime, speed, entered state, and so forth. The output, represented in XML files, is used as input for GNNs of the proposed system.

In the proposed system, charging stations are treated as special edges. The location information of the charging stations is extracted using the Open Charge Map API [10] for the area of interest. Based on the longitude and latitude, charging stations were assigned to the respective edges. Relevant features such as power, cost, and number of points were extracted from the Open Charge Map API and assigned to charging stations.

The proposed system was implemented using Python 3.13.7. The major packages used to demonstrate the proposed tourism planning system were `gymnasium` and `traci`. The `gymnasium` package was utilized to develop the RL algorithm, and the `traci` package was utilized for controlling vehicles' routing behavior.

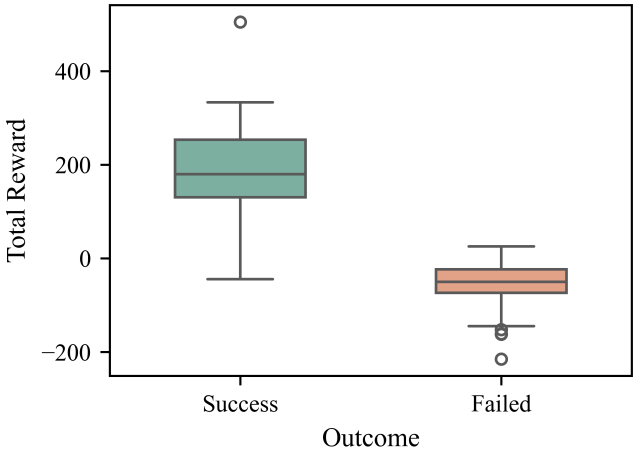


Fig. 2: Reward Distribution

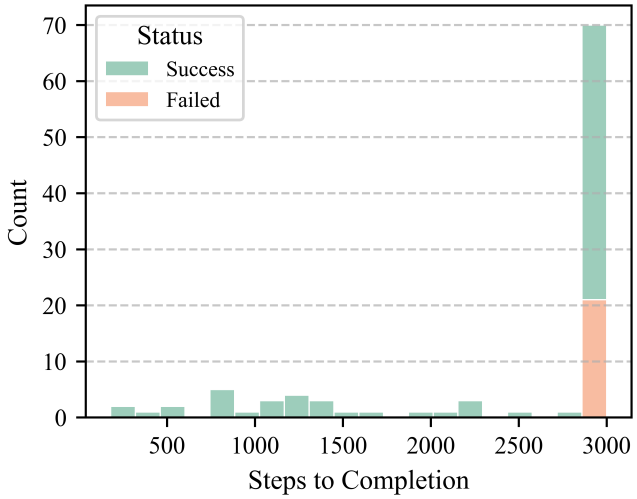


Fig. 3: Trip Duration for Successful and Failed Experiments

B. Energy and Travel Time Prediction

In experiments, the digraph created from the Kochi location of India consisted of 52,418 nodes and 1,19,861 edges. A GNN was constructed on top of this graph with edge weights. And a custom RL agent was developed on top of the GNN implementation to interact with the tourism planning environment. Accordingly, it was able to choose the next travel spot considering constraints such as battery availability, charging station location, traffic state, elevation conditions, and so forth. The table I shows the GNN predicted energy in kWh and travel time values of destinations in seconds.

C. Success Rate and RL Rewards

To evaluate the success rate and rewards obtained by the proposed RL agents, we performed 100 experiments – i.e.,

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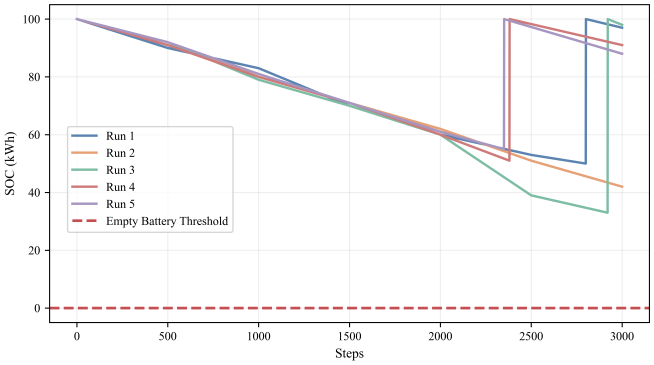


Fig. 4: Battery Utilization & Recharge

100 different travel trips were chosen in a Kochi location map of India. The travel trips represent a random start and random destination points. Among 100 different experiments, the proposed GNN-RL was capable of achieving a 79% success rate. In fact, a few modern algorithms, including transformer architectures, could probably provide a better success rate. However, such architectures could lead to hefty computational overheads, limiting the utility in real-world situations.

Figure 2 depicts the reward distribution for success and failure cases. For successful situations, we observed that the RL agent had an average positive reward of 180, whereas the failure cases were accumulating an average negative reward of -60. And this is a clear indication that the RL agent was confident in executing successful cases. But the variance of success cases from 0 to 500 indicated that the agent is taking diverse actions to reach the destination.

D. Travel Duration and Impacts

The travel trip duration for both the success and failure random experiments is shown in Figure 3. The graph reveals that there were around 30% of trips that had fewer than 10 steps; around 50% of successful trips have a step length of 3000 steps.

From Figure 3, it is possible to observe that some trips took long routes to reach the destination. Also, all failed experiments took over 3000 steps because the environment was trying to find feasible paths to reach the destination before they failed.

E. Battery Behavior Analysis

To study the behavior of batteries of EVs, five experimental runs were performed using the proposed GNN-RL system. It was observed that the battery of EVs kept draining over multiple steps. However, the RL agent pointed out a few recharging stations, and the EVs were able to charge the batteries. To do so, the RL agent kept rerouting the vehicle to the nearest optimal charging station, considering time and location.

Figure 4 depicts the scenario, pointing out the charging situations of EVs in a number of steps. For instance, run 3 waited almost 2850 steps to identify the optimal charging station when compared with run 4. Once the recharging of the batteries of EVs happens at the intended step, the batteries of EVs return back to nearing 100%.

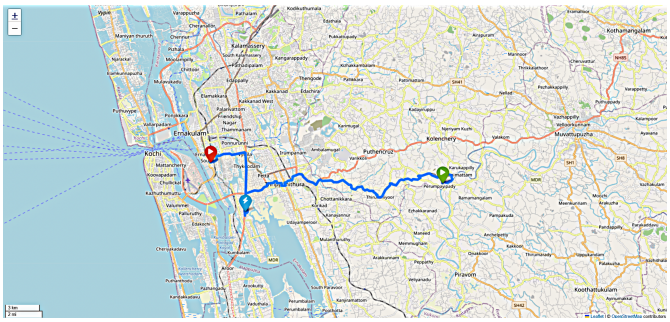


Fig. 5: Predicted Route – Map Integration

F. Map Integration

Figure 5 depicts the path from the starting location marked in green to the destination location marked in red. The proposed GNN-RL approach identified the intermediate steps of a tourism location considering the remaining battery capacity to reach the destination. It could be observed that the EV had to take multiple paths by identifying an optimal charging station. Notably, the vehicle deviated from its path and re-routed to an optimal charging station selected by the RL agent, and the same was marked in blue.

The conducted experiments and validation results showed that the proposed system predicted the optimal path with a success rate of 79%. The approx_kl value of 0.608 indicated that the PPO model was able to learn the environment really well, and the positive reward of 57 indicated that the system was able to reach the destination with the battery charging criterion. The system was taking a mean length of 2700 to reach the destination in the experiment.

V. Conclusion and Future Work

Electric vehicles increased, and associated research directions magnified, owing to the core interests of addressing the climate change conditions of the globe. Most importantly, the tourism sector of various countries has opted to keenly identify novel solutions to maintain ecological requirements and improve the green index of nations.

This article proposed a GNN-RL-based travel plan for EV travelers, considering various complex parameters, including the battery status of EVs, in real-time. The proposed mechanism included a few real-time map-related APIs and RL agents to quickly identify the travel plan while EV travelers are on the way to enjoying tours. Experiments, conducted at the IoT Cloud research laboratory, were carried out, and the findings were reported in the article. Most importantly, the final integration of the solution to maps revealed a success rate of 79% for 100 different random experiments performed in a tour location of Kochi in India.

In the future, the approach could include a few other complex parameters such as weather and traffic conditions of a tour location.

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